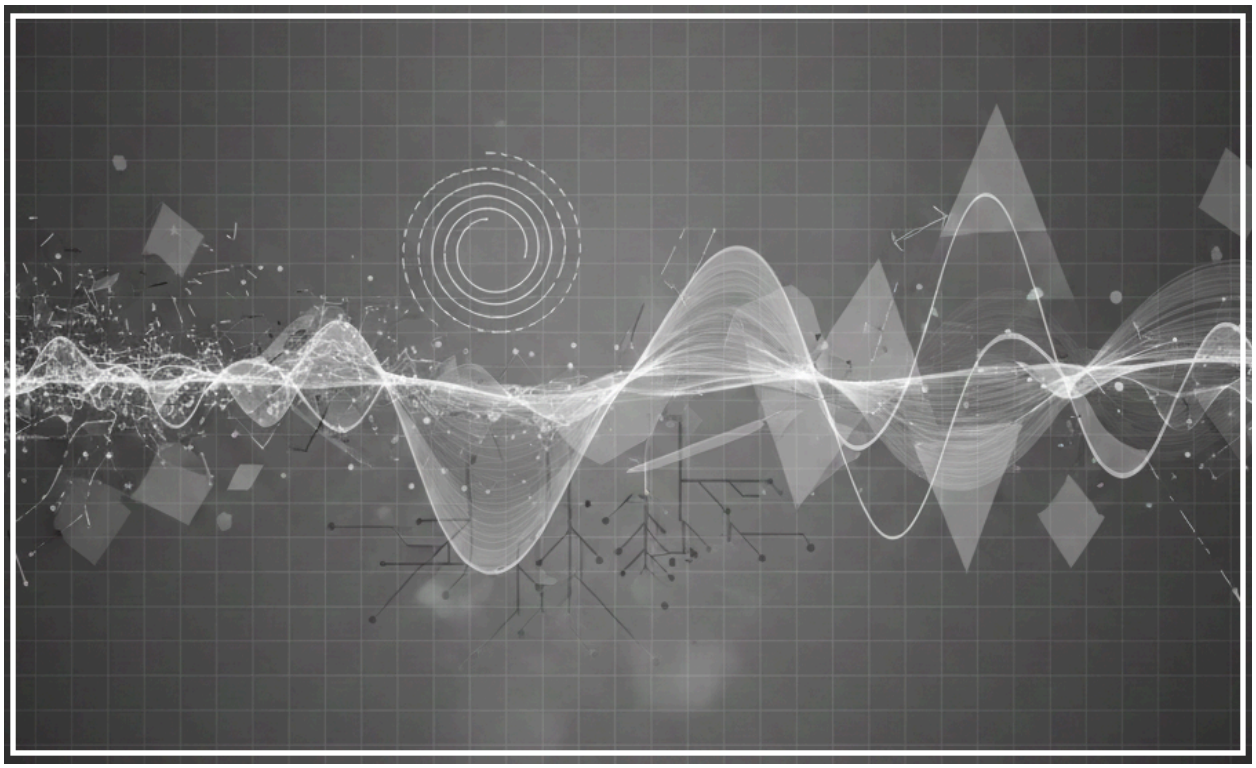
Assignment 1 (Optional Bonus: TimesFM Forecast)

DATA.STAT.780 Time Series Analysis (Spring 2026, TAU)



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MSc in Data Science

Column Assigned: V87

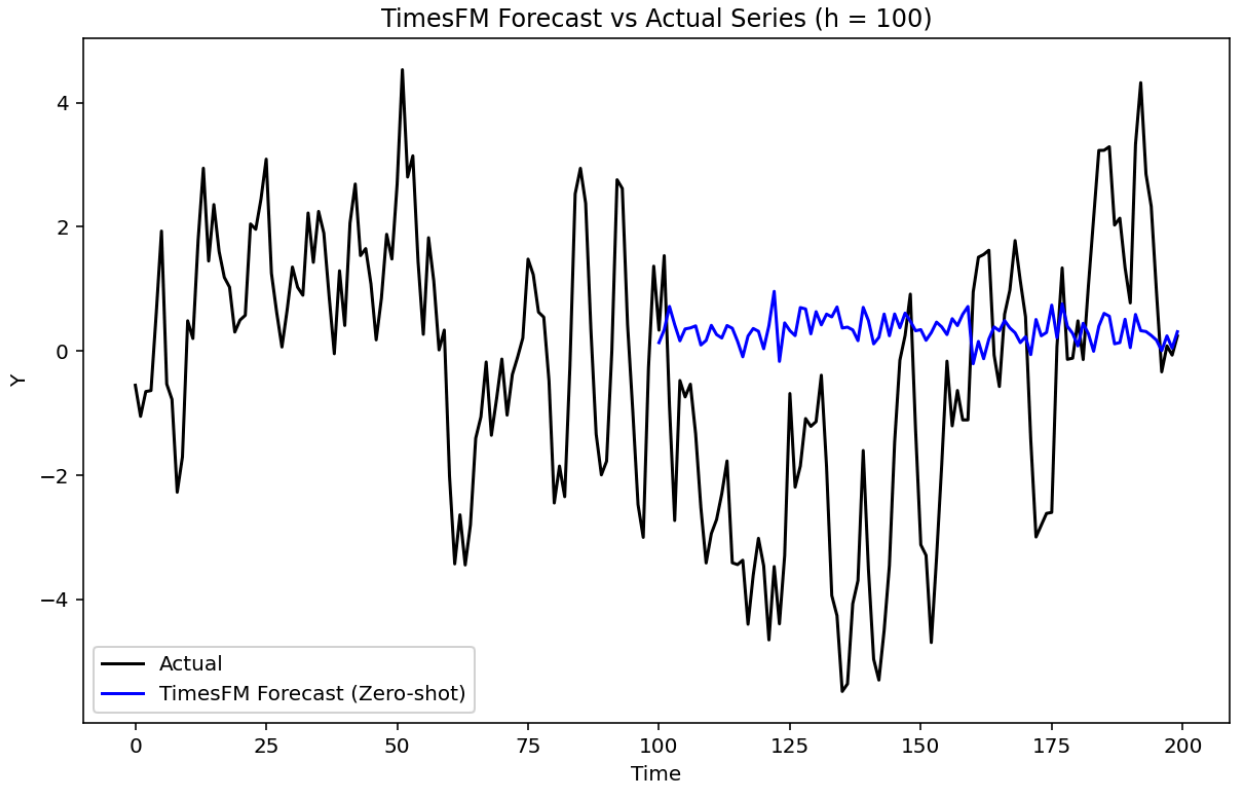
Optional Bonus: TimesFM Forecast (Compare with ARIMA)

Python Code:

```python
# Student Name: Anas Uddin
# Student ID: 153215593
# Assigned Column: V87
# Bonus: TimesFM Forecast (Compare with ARIMA)
# ---------------------------------------------

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Import PyTorch TimesFM
from timesfm import TimesFM_2p5_200M_torch as TimesFmTorch
from timesfm import ForecastConfig

# Load Data
data = pd.read_csv("Case_study.csv")
Y = data["V87"].values

# Split into training (1-100) and test (101-200)
Y_train = Y[:100]
Y_test = Y[100:]

# Pad Y_train to multiple of patch size
patch_size = 32  # default patch size for TimesFM_2p5_200M_torch
pad_len = (patch_size - (len(Y_train) % patch_size)) % patch_size
Y_train_padded = np.concatenate([Y_train, np.zeros(pad_len)])

# Initialize TimesFM Model
model = TimesFmTorch()

# Set max_horizon = 100
forecast_config = ForecastConfig(max_horizon=100)

# Compile the model
model.compile(forecast_config=forecast_config)

# Zero-Shot Forecast
```

```
forecast_zero = model.forecast(inputs=[Y_train_padded], horizon=100)
pred_zero = forecast_zero[0][: len(Y_test)]  # Trim extra padded points

# Compute MSE
mse_zero = np.mean((Y_test - pred_zero) ** 2)
print("Zero-shot MSE:", mse_zero)

# Plot Actual vs Forecast
plt.figure(figsize=(10, 6))

# Actual series
plt.plot(range(200), Y, label="Actual", color="black")

# Zero-shot forecast (flattened)
plt.plot(
    range(100, 200),
    pred_zero.flatten(),
    label="TimesFM Forecast (Zero-shot)",
    color="blue",
)

plt.xlabel("Time")
plt.ylabel("Y")
plt.title("TimesFM Forecast vs Actual Series (h = 100)")
plt.legend()
plt.show()
```

Output:



I applied the pretrained TimesFM (2.5–200M PyTorch version) model to forecast the next 100 observations (h = 100) using the first 100 observations as training data. The zero-shot TimesFM forecast produced:

- TimesFM MSE = 7.5321
- ARIMA MSE = 7.4201

Comparing the two models, ARIMA achieved a slightly lower MSE than TimesFM. Although the difference is small, ARIMA performs marginally better for this particular time series.

From the forecast plot, TimesFM successfully captures the general level and trend of the series but produces smoother forecasts. ARIMA appears to better capture short-term fluctuations and variability in the data.

Overall, for my dataset, the classical ARIMA model slightly outperforms the foundation model in terms of forecasting accuracy.